

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Designed and maintained **gRPC inter-service layer**; resolved production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing – **exactly-once delivery at sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime, 44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Led backend and distributed systems development for a 3-person engineering team – designed **fault-tolerant production systems** end-to-end using Python, gRPC, PostgreSQL, Redis, and AWS infrastructure
- Architected **vLLM inference engine with PagedAttention** – designed distributed compute scheduling for concurrent LLM requests; **zero OOM failures** at production traffic with continuous batching

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users at sub-50ms p99, 10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; shipped to **3B+ active users**
- Collaborated with Chrome infrastructure team to deliver design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Kubernetes, Multi-AZ, Event-driven architecture

Languages: C++, Go, Python, Rust, Java, TypeScript, JavaScript, SQL, Bash

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM APIs (Claude, GPT), LangChain, LLM-as-a-judge evaluation, PyTorch

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor